

MAT 524: Functions of a Real Variable II

Homework III

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[[Problem No. 1]]:

Problem Statement

Suppose $f_n, f \in L^1(E)$. We say that $f_n \rightarrow f$ in L^1 if $\|f - f_n\|_1 \rightarrow 0$, i.e.,

$$\int_E |f_n - f| \rightarrow 0.$$

Compare the two conditions

(a) $f_n \rightarrow f$ almost everywhere

(b) $f_n \rightarrow f$ in L^1 .

Determine whether (a) implies (b) or (b) implies (a). Give a proof or counter-example for each implication.

Solution

Counter-example for (a) $\Rightarrow$ (b): Suppose $E = [0, 1]$ and $f_n(x) = nx^2(1 - x^3)^n$. For $x \in E$,

$$\lim_{n \rightarrow \infty} nx^2(1 - x^3)^n = 0$$

because a^n shrinks faster than n grows for $0 \leq a < 1$. So $f_n \rightarrow 0$ almost everywhere, but

$$\lim_{n \rightarrow \infty} \int_E |nx^2(1 - x^3)^n - 0| dx = \lim_{n \rightarrow \infty} \frac{-n}{3} \int_E -3x^2(1 - x^3)^n dx \tag{1}$$

$$= \lim_{n \rightarrow \infty} \left[\frac{-n(1 - x^3)^{n+1}}{3n + 3} \right]_0^1 \tag{2}$$

$$= \frac{1}{3}. \tag{3}$$

Thus, f_n doesn't converge to f in L^1 .

Counter-example for (b) $\Rightarrow$ (a): Suppose $E = [0, 1]$ and $f_n = \sin(nx)$. Note that f_n converges to 0 in L^1 because

$$\lim_{n \rightarrow \infty} \int_E |\sin(nx) - 0| \, dx = \lim_{n \rightarrow \infty} \int_E \sin(nx) \, dx \quad (4)$$

$$= \lim_{n \rightarrow \infty} \left[\frac{-\cos(n)}{n} + \frac{1}{n} \right] \quad (5)$$

$$= 0. \quad (6)$$

But $f_n = \sin(nx)$ does not converge a.e. on E as $n \rightarrow \infty$, due to the oscillating nature of the sine function. ■



[[Problem No. 2]]:

Problem Statement

Let f_n, f be integrable measurable functions on E . Assume $f_n \rightarrow f$ almost everywhere. Suppose that

$\lim_{n \rightarrow \infty} \int_E |f_n| = \int_E |f|$. Prove that $f_n \rightarrow f$ in $L^1(E)$.

Solution

By the Triangle Inequality, $|f_n - f| \leq |f_n| + |f|$ for all $n \in \mathbb{N}$. Furthermore, since $\lim_{n \rightarrow \infty} \int_E |f_n| = \int_E |f|$ and f is integrable, we have

$$\lim_{n \rightarrow \infty} \int_E |f_n| + \int_E |f| = \int_E |f| + \int_E |f|, \quad (7)$$

i.e.,

$$\lim_{n \rightarrow \infty} \int_E (|f_n| + |f|) = 2 \int_E |f| \quad (8)$$

$$< \infty. \quad (9)$$

Thus, by the Generalized Dominated Convergence Theorem (GDCT),

$$\lim_{n \rightarrow \infty} \int_E |f_n - f| = \int_E \lim_{n \rightarrow \infty} |f_n - f| \quad (10)$$

$$= 0. \quad (11)$$

Therefore, f_n converges to f in $L^1(E)$.

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[[Problem No. 3]]:

Problem Statement

Let $C([0, 1])$ denote the space of continuous real-valued functions on the interval $[0, 1]$. Show that it is complete relative to the max-norm.

Solution

Let $\{f_n\}$ be a Cauchy sequence of continuous functions in $C([0, 1])$. Since $C([0, 1])$ is a subspace of $L^\infty([0, 1])$ and $[0, 1]$ is measurable, we can use Proposition 5 and Theorem 7 to conclude that $\{f_n\}$ has a rapidly Cauchy subsequence that converges. Thus, by Proposition 4, $\{f_n\}$ converges. Let us call the function to which $\{f_n\}$ converges f . To show that $C([0, 1])$ is complete with respect to the max-norm, we must show that $f \in C([0, 1])$. In order to achieve this, we will show that $\|f_n - f\|_\infty \rightarrow 0$.

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[[Problem No. 4]]:

Problem Statement

Let f be a bounded measurable function on $[0, 1]$. Prove that

$$\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty.$$

Solution

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[[Problem No. 5]]:

Problem Statement

Let $1 \leq p < \infty$. Suppose $f_n, f \in L^p(E)$ and $f_n \rightarrow f$ pointwise almost everywhere. Show that $\|f_n\|_p \rightarrow \|f\|_p$ if and only if $f_n \rightarrow f$ in $L^p(E)$.

Solution

For the reverse direction of the proof, we need to use the p -norm analogue of the reverse triangle inequality.

We have the following lemma:

Lemma 5.1: If f, g are in $L^p(E)$, then $\left| \|f\|_p - \|g\|_p \right| \leq \|f - g\|_p$.

Proof: By Minkowski's Inequality, we have

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Using this, we obtain the following inequalities:

$$\|f\|_p = \|f - g + g\|_p \leq \|f - g\|_p + \|g\|_p \tag{12}$$

$$\|g\|_p = \|g - f + f\|_p \leq \|g - f\|_p + \|f\|_p \tag{13}$$

Rearranging (12) and (13), we see that

$$\|f\|_p - \|g\|_p \leq \|f - g\|_p \tag{14}$$

$$\|g\|_p - \|f\|_p \leq \|g - f\|_p. \tag{15}$$

But $\|f - g\|_p = \|g - f\|_p$, so we conclude that

$$\left| \|f\|_p - \|g\|_p \right| \leq \|f - g\|_p.$$

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Now we can move on the the proof of the result in the problem statement.

Proof of Main Result: ($\Rightarrow$) First suppose that $\|f_n\|_p \rightarrow \|f\|_p$. Then

$$\lim_{n \rightarrow \infty} \left[\int_E |f_n|^p \right]^{\frac{1}{p}} = \left[\int_E |f|^p \right]^{\frac{1}{p}} \quad (16)$$

Taking the p th power of both sides of Equation (16) and applying the limit definition of continuity, we see that

$$\lim_{n \rightarrow \infty} \int_E |f_n|^p = \int_E |f|^p. \quad (17)$$

Adding $\int_E |f|^p$ (which is finite, since $f \in L^p(E)$) to both sides of Equation (17) and multiplying the result by 2^p , we obtain

$$\lim_{n \rightarrow \infty} \int_E 2^p (|f_n|^p + |f|^p) = 2^{p+1} \int_E |f|^p \quad (18)$$

$$< \infty. \quad (19)$$

We also have that $|f_n - f|^p \leq 2^p (|f_n|^p + |f|^p)$. Thus, by the GDCT,

$$\lim_{n \rightarrow \infty} \left[\int_E |f_n - f|^p \right]^{\frac{1}{p}} = \left[\lim_{n \rightarrow \infty} \int_E |f_n - f|^p \right]^{\frac{1}{p}} \quad (20)$$

$$= \left[\int_E \lim_{n \rightarrow \infty} |f_n - f|^p \right]^{\frac{1}{p}} \quad (21)$$

$$= 0^{\frac{1}{p}} \quad (22)$$

$$= 0. \quad (23)$$

Therefore, $f_n \rightarrow f$ in $L^p(E)$.

($\Leftarrow$) Next, suppose that $f_n \rightarrow f$ in $L^p(E)$. By Lemma 5.1,

$$\left| \|f_n\|_p - \|f\|_p \right| \leq \|f_n - f\|_p. \quad (24)$$

Taking the limit on both sides of (24) yields

$$\lim_{n \rightarrow \infty} \left| \|f_n\|_p - \|f\|_p \right| \leq 0, \quad (25)$$

since $f_n \rightarrow f$ in $L^p(E)$. But the right-hand side of (25) must be non-negative, so it follows that

$$\lim_{n \rightarrow \infty} \left| \|f_n\|_p - \|f\|_p \right| = 0, \quad (26)$$

i.e.,

$$\lim_{n \rightarrow \infty} \|f_n\|_p = \|f\|_p, \quad (27)$$

as desired. ■

